ANFIS: Adaptive Neuro-Fuzzy Inference Systems

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Summary

- Introduction
- ANFIS Architecture
- Hybrid Learning Algorithm
- ANFIS as a Universal Approximatior
- Simulation Examples

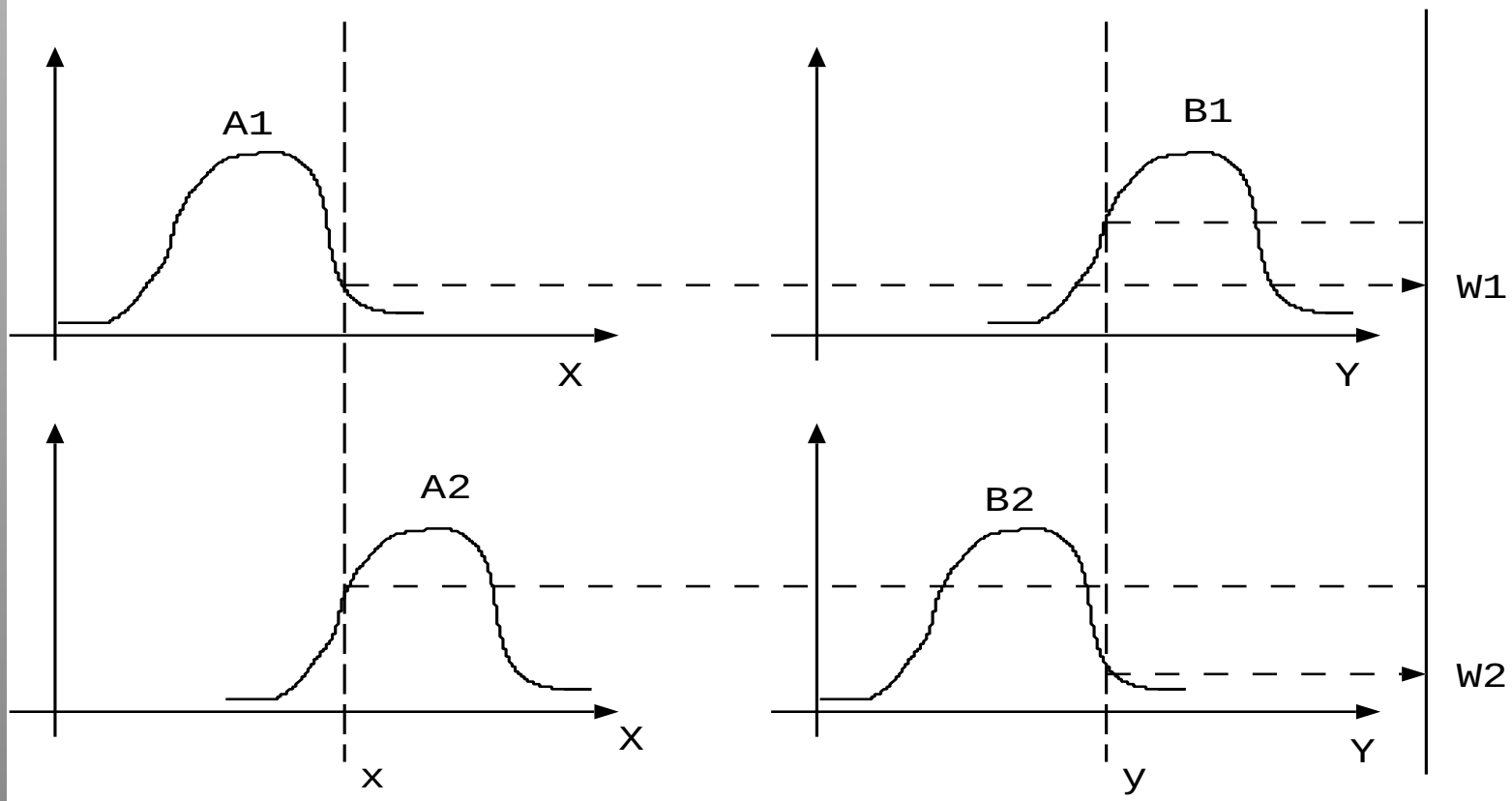
Introduction

- ANFIS: Artificial Neuro-Fuzzy Inference Systems
- ANFIS are a class of adaptive networks that are funcionally equivalent to fuzzy inference systems.
- ANFIS represent Sugeno e Tsukamoto fuzzy models.
- ANFIS uses a hybrid learning algorithm

Sugeno Model

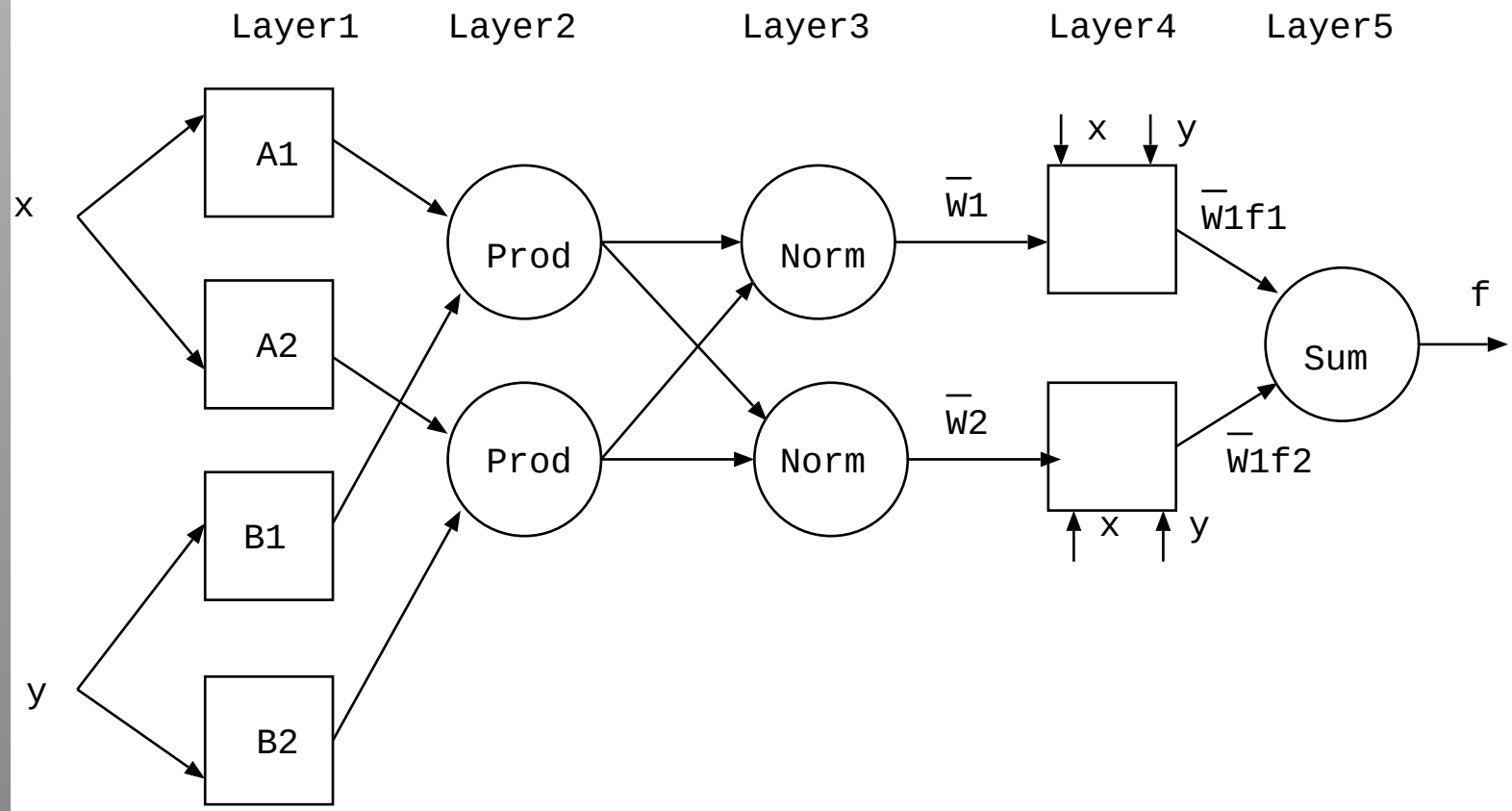
- Assume that the fuzzy inference system has two inputs x and y and one output z.
- A first-order Sugeno fuzzy model has rules as the following:
- Rule1:
  If x is A_1 and y is B_1, then $f_1 = p_1 x + q_1 y + r_1$
- Rule2:
  If x is A_2 and y is B_2, then $f_2 = p_2 x + q_2 y + r_2$

Sugeno Model - I



$f1=p1x+q1y+r1$ $f2=p2x+q2y+r2$ $f=\frac{w1.f1+w2.f2}{w1+w2}$

ANFIS Architecture



Layer 1 - I

- $O_{l,i}$ is the output of the ith node of the layer l.
- Every node i in this layer is an adaptive node with a node function
  $O_{1,i} = \mu_{A_i}(x)$ for $i = 1, 2$, or
  $O_{1,i} = \mu_{B_{i-2}}(x)$ for $i = 3, 4$
- x (or y) is the input node i and A_i (or B_{i-2}) is a linguistic label associated with this node
- Therefore $O_{1,i}$ is the membership grade of a fuzzy set (A_1, A_2, B_1, B_2).

Layer 1 - II

- Typical membership function:

$$\mu_A(x) = \frac{1}{1 + \left|\frac{x - c_i}{a_i}\right|^{2b_i}}$$

- a_i, b_i, c_i is the parameter set.
- Parameters are referred to as **premise parameters**.

Layer 2

- Every node in this layer is a fixed node labeled `Prod`.
- The output is the product of all the incoming signals.
- $O_{2,i} = w_i = \mu_{A_i}(x) \cdot \mu_{B_i}(y), \quad i = 1, 2$
- Each node represents the fire strength of the rule
- Any other T-norm operator that perform the AND operator can be used

Layer 3

- Every node in this layer is a fixed node labeled `Norm`.
- The ith node calculates the ratio of the ith rulet's firing strenght to the sum of all rulet's firing strengths.
- $O_{3,i} = \overline{w}_i = \frac{w_i}{w_1 + w_2}, \quad i = 1, 2$
- Outputs are called **normalized firing strengths**.

Layer 4

- Every node i in this layer is an adaptive node with a node function:

$$O_{4,1} = \overline{w}_i f_i = \overline{w}_i (p_x + q_i y + r_i)$$

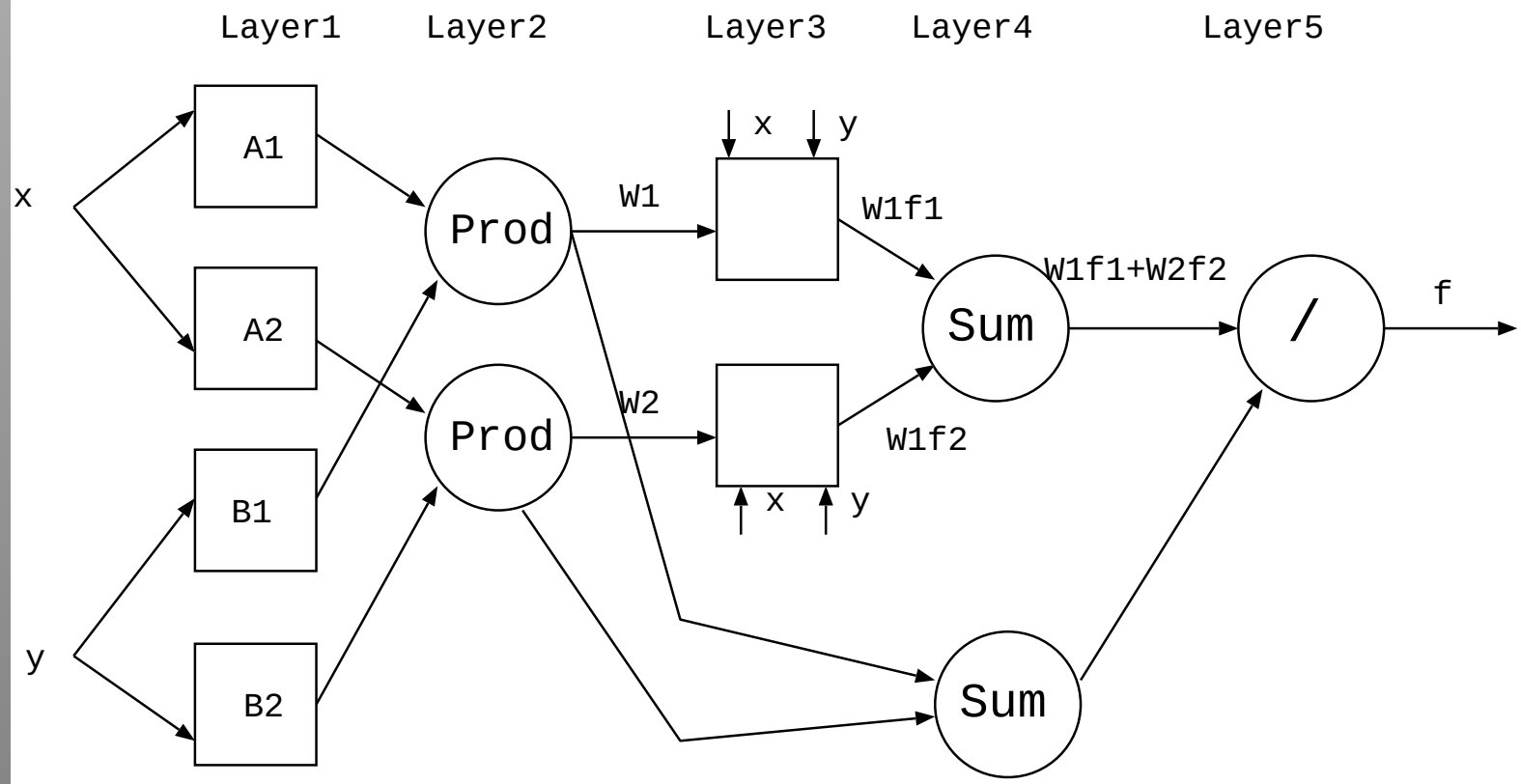
- $\overline{w}_i$ is the normalized firing strenght from layer 3.
- $\{p_i, q_i, r_i\}$ is the parameter set of this node.
- These are referred to as **consequent parameters**.

Layer 5

- The single node in this layer is a fixed node labeled sum, which computes the overall output as the summation of all incoming signals:
- $overall\ output = O_{5,1} = \sum_i \overline{w}_i f_i = \frac{\sum_i w_i f_i}{\sum_i w_i}$

Alternative Structures

- There are other structures



Learning Algorithm

Hybrid Learning Algorithm - I

- The ANFIS can be trained by a hybrid learning algorithm presented by Jang in the chapter 8 of the book.
- In the forward pass the algorithm uses least-squares method to identify the consequent parameters on the layer 4.
- In the backward pass the errors are propagated backward and the premise parameters are updated by gradient descent.

Hybrid Learning Algorithm - II

	Forward Pass	Backward Pass
Premise Parameters	Fixed	Gradient Descent
Consequent Parameters	Least-squares estimator	Fixed
Signals	Node outputs	Error signals

Two passes in the hybrid learning algorithm for ANFIS.

Universal Aproximator

ANFIS is a Universal Aproximator

- When the number of rules is not restricted, a zero-order Sugeno model has unlimited approximation power for matching any nonlinear function arbitrarily well on a compact set.
- This can be proved using the Stone-Weierstrass theorem.
- Let D be a compact space of N dimensions, and let F be a set of continuous real-valued functions on D satisfying the following criteria:

Stone-Weierstrauss theorem - I

- Let D be a compact space of N dimensions, and let F be a set of continuous real-valued functions on D satisfying the following criteria:

Indentity function: The constant $f(x) = 1$ is in F.

Separability: For any two points $x_1 \neq x_2$ in D, there is an f in F such that $f(x_1) \neq f(x_2)$.

Algebraic closure: If f and g are any two functions in F, then fg and $af + bg$ are in F for any two real numbers a and b.

Stone-Weierstrauss theorem - II

- Then F is dense on $C(D)$, the set of continuous real-valued functions on D.
- For any $\epsilon > 0$ and any function g in $C(D)$, there is a function f in F such that $|g(x) - f(x)| < \epsilon$ for all $x \in D$.
- The ANFIS satisfies all these requirements.

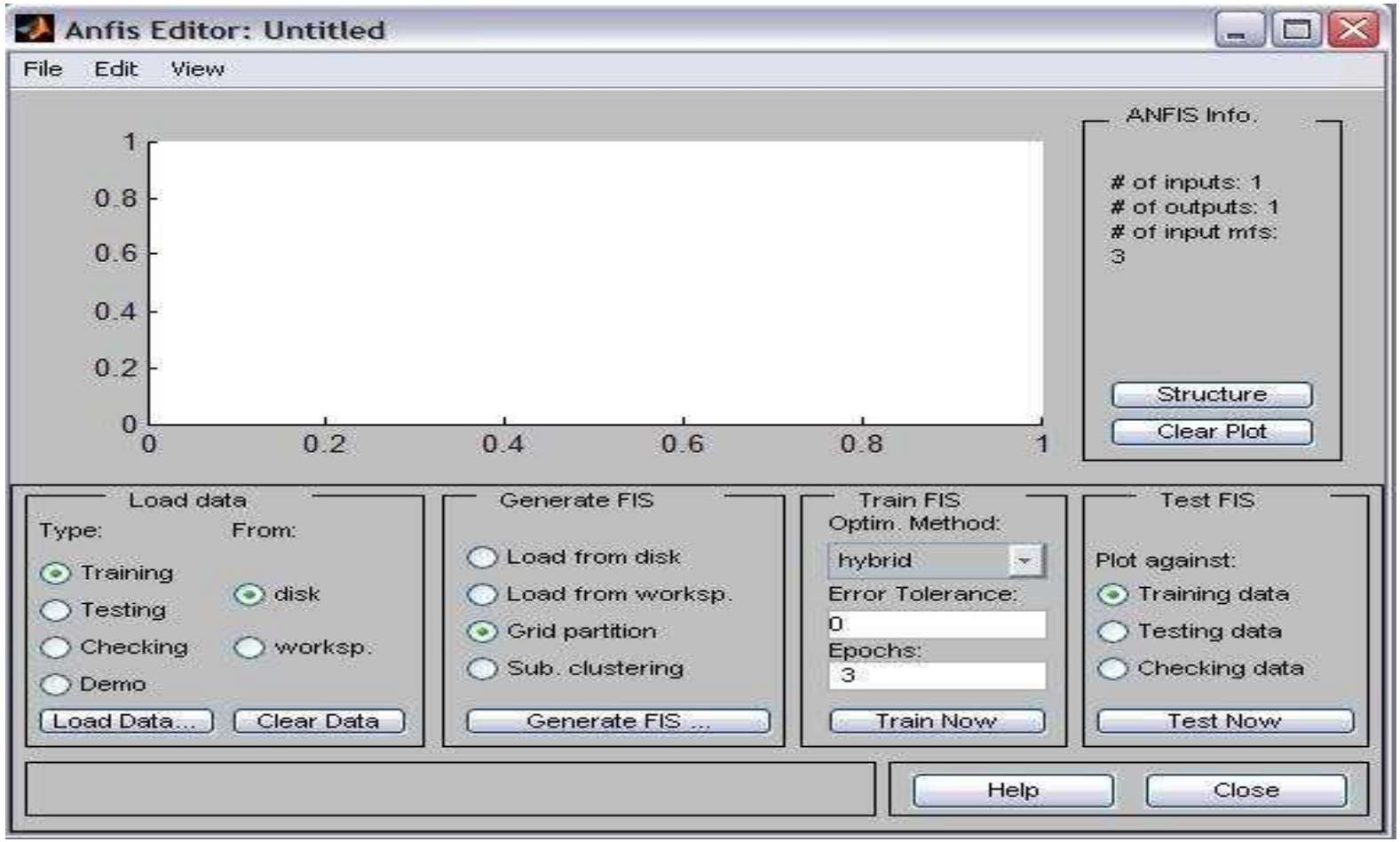
Anfis and Matlab

Matlab

- It is possible to use a graphics user interface
- Command `anfisedit`.

- It is possible to use the command line interface or m-file programs.
- There are functions to generate, train, test and use these systems.

ANFIS gui


Anfis Editor: Untitled
File
Edit
View
1
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1
ANFIS Info.
of inputs: 1
of outputs: 1
of input mfs:
3
Structure
Clear Plot
Load data
Type:
From:
Training
Testing
Checking
Demo
disk
worksp.
Load Data...
Clear Data
Generate FIS
Load from disk
Load from worksp.
Grid partition
Sub. clustering
Generate FIS ...
Train FIS
Optim. Method:
hybrid
Error Tolerance:
0
Epochs:
3
Train Now
Test FIS
Plot against:
Training data
Testing data
Checking data
Test Now
Help
Close

Applying

- Initializing
- Training
- Testing
- Using

Initializing - GENFIS1 - 1

- `FIS = GENFIS1(DATA)` generates a single-output Sugeno-type fuzzy inference system (`FIS`) using a grid partition on the data (no clustering).
- `FIS` is used to provide initial conditions for posterior ANFIS training.
- `DATA` is a matrix with N+1 columns where the first N columns contain data for each `FIS` input, and the last column contains the output data.

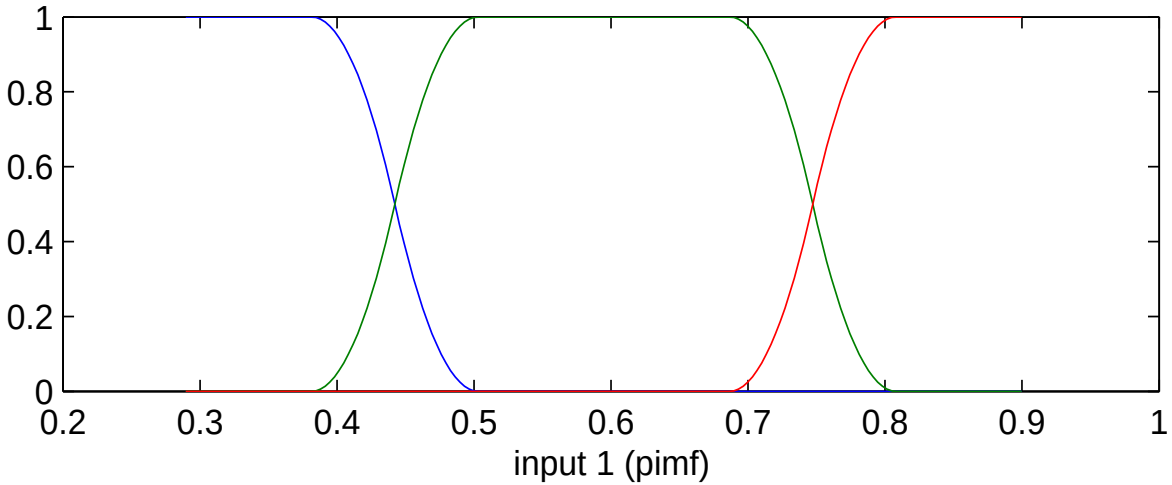
Initializing - GENFIS1 - 2

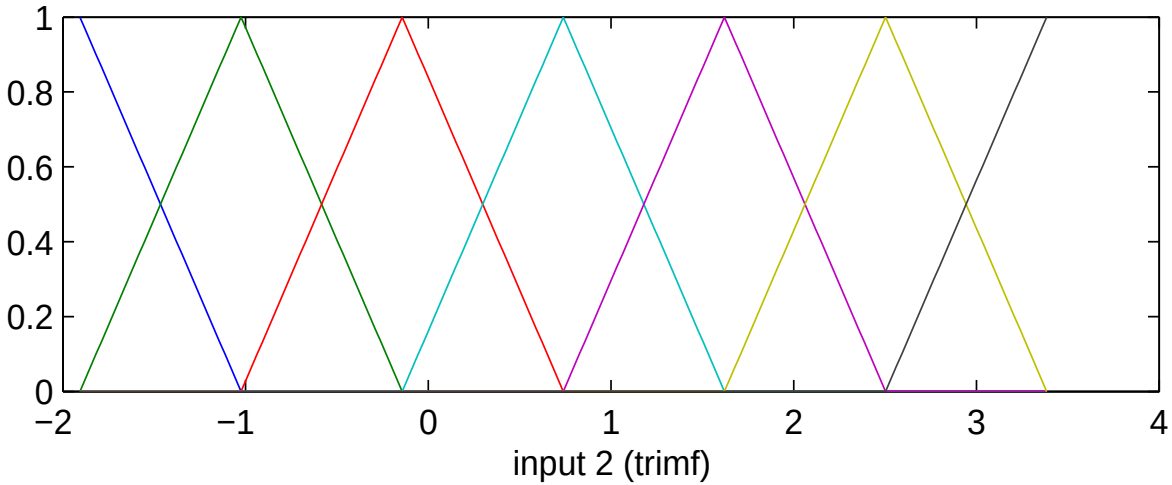
- By default `GENFIS1` uses two `'gbellmf'` type membership functions for each input.
- Each rule generated has one output membership function, which is of type 'linear' by default.
- It is possible to define these parameters using `FIS = GENFIS1(DATA, NUMMFS, INPUTMF, OUTPUTMF)`
- `fis = genfis1(data, [3 7], char('pimf', 'trimf'));`

Initializing - GENFIS1 - 3

```
data = [rand(10,1) 10*rand(10,1)-5 rand(10,1)];
fis = genfis1(data, [3 7], char('pimf','trimf'));
[x,mf] = plotmf(fis,'input',1);
subplot(2,1,1), plot(x,mf);
xlabel('input 1 (pimf)');
[x,mf] = plotmf(fis,'input',2);
subplot(2,1,2), plot(x,mf);
xlabel('input 2 (trimf)');
```

Initializing - GENFIS1 - 4





Initializing - GENFIS2

- `GENFIS2` generates a Sugeno-type FIS using subtractive clustering.
- `GENFIS2` extracts a set of rules that models the data behavior.
- The rule extraction method first determines the number of rules and antecedent membership functions and then uses linear least squares estimation to determine each rule's consequent equations.

Training

- `ANFIS` uses a hybrid learning algorithm to identify the membership function parameters of single-output, Sugeno type fuzzy inference systems (FIS).
- There are many ways of using this function.
- Some examples:
  - `[FIS,ERROR] = ANFIS(TRNDATA)`
  - `[FIS,ERROR] = ANFIS(TRNDATA,INITFIS)`

Using

- `EVALFIS` evaluates a fuzzy inference system.
- `Y = EVALFIS(U,FIS)` simulates the Fuzzy Inference System FIS for the input data U and returns the output data Y.

Example

- run exemplo06_03.m

The End